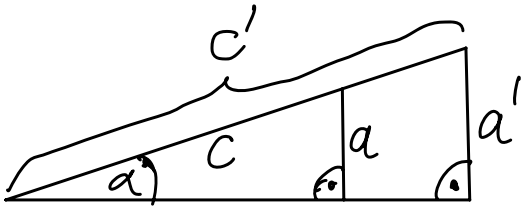
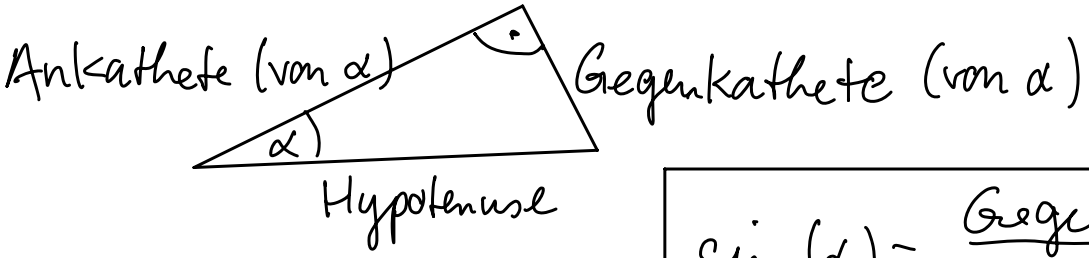
7 Trigonometrie

Rechtwinkliges Dreieck:



Ähnlichkeit $\Rightarrow \frac{a}{c} = \frac{a'}{c'}$

Zu gegebenem Winkel α hat dieses Verhältnis immer denselben Wert, der als $\sin(\alpha)$ bezeichnet wird.



Sinus, Cosinus, Tangens am rechtwinkligen Dreieck

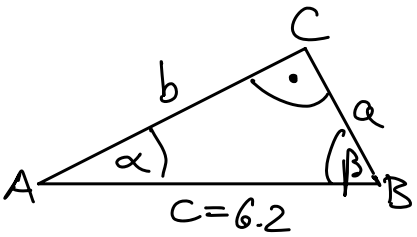
$$\sin(\alpha) = \frac{\text{Gegenkath.}}{\text{Hypotenuse}}$$

$$\cos(\alpha) = \frac{\text{Ankath.}}{\text{Hypotenuse}}$$

$$\tan(\alpha) = \frac{\text{Gegenkath.}}{\text{Ankath.}}$$

Ü7.1 γ ist der rechte Winkel!

a) $c = 6.2$, $\alpha = 34^\circ$



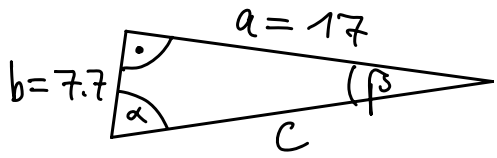
$$\sin(34) = \frac{a}{6.2}$$

$$\Rightarrow a = 6.2 \cdot \sin(34) = 3.47$$

$$\cos(34) = \frac{b}{6.2} \Rightarrow b = 6.2\cos(34) = 5.14$$

$$\beta = 90 - \alpha = 56^\circ$$

b) $a = 17, \quad b = 7.7$



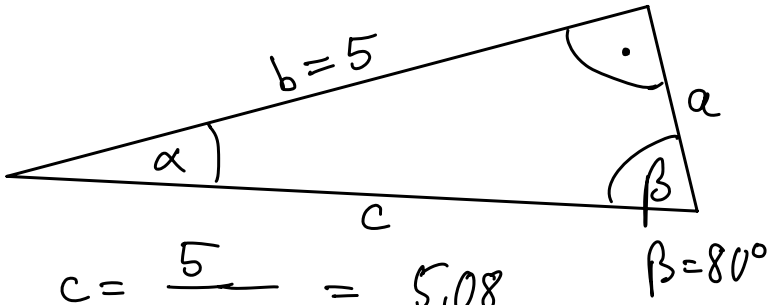
$$\tan(\alpha) = \frac{17}{7.7} \Rightarrow \alpha = \tan^{-1}\left(\frac{17}{7.7}\right) = 65.63^\circ$$

$$\Rightarrow \beta = 90 - \alpha = 24.37^\circ$$

$$\sin(\alpha) = \frac{17}{c} \Rightarrow c = \frac{17}{\sin(65.63)} = 18.66$$

c)

$$\alpha = 90 - \beta = 10^\circ$$

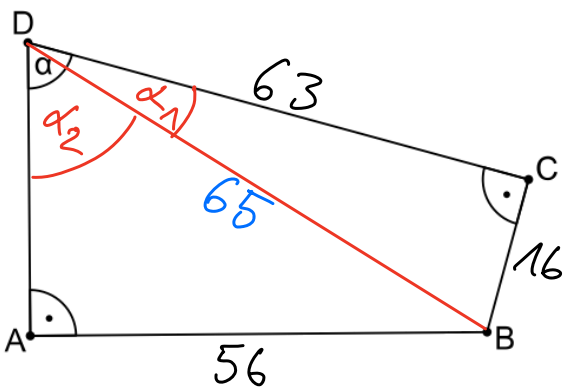


$$\sin(80) = \frac{5}{c} \Rightarrow c = \frac{5}{\sin(80)} = 5.08$$

$$\tan(80) = \frac{5}{a} \Rightarrow a = \frac{5}{\tan(80)} = 0.88$$

Ü 7.2 $AB = 56, \quad BC = 16$

$CD = 63, \quad \alpha = ?$



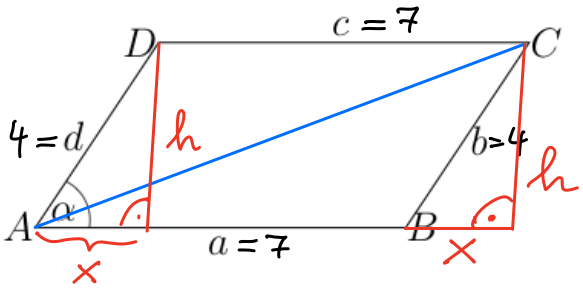
$$\tan(\alpha_1) = \frac{16}{63} \Rightarrow \alpha_1 = 14.25^\circ$$

$$BD = \sqrt{16^2 + 63^2} = 65$$

$$\sin(\alpha_2) = \frac{56}{65} \Rightarrow \alpha_2 = 59.49^\circ \Rightarrow \alpha = \alpha_1 + \alpha_2 = 73.74^\circ$$

Ü 7.3 Parallelogramm ABCD

$a = 7,\ b = 4,\ \alpha = 50°$



Fläche = ? Diagonale AC = ?

$d = b = 4, \quad a = c = 7$

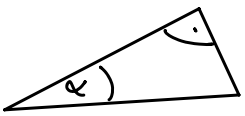
$\sin(50) = \frac{h}{4} \Rightarrow h = 4 \cdot \sin(50) = 3.0642$ cm

Fläche $= a \cdot h = 21.45$ cm^2

$\cos(50) = \frac{x}{4} \Rightarrow x = 4 \cdot \cos(50) = 2.5712$

$AC^2 = (a+x)^2 + h^2 \Rightarrow AC = 10.0497$

Bisher ist z.B. $\sin(\alpha)$ nur definiert für $0 < \alpha < 90°$



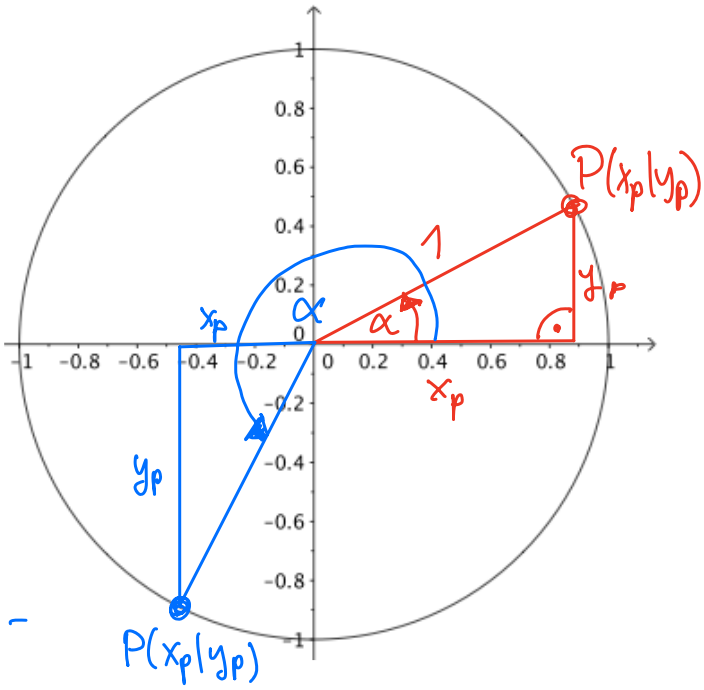
$\sin(\alpha) = \frac{\text{Gegenkath.}}{\text{Hyp.}}$ (am rechtw. Dreieck ist $\alpha < 90°$)

Am Einheitskreis ($r = 1$, $M(0/0)$) ergibt das, zunächst für $0 < \alpha < 90°$, dass die Koordinate von P durch

$\sin(\alpha) = \frac{y_P}{1}, \quad \cos(\alpha) = \frac{x_P}{1}$

$\Rightarrow x_P = \cos(\alpha), \quad y_P = \sin(\alpha)$

gegeben sind.



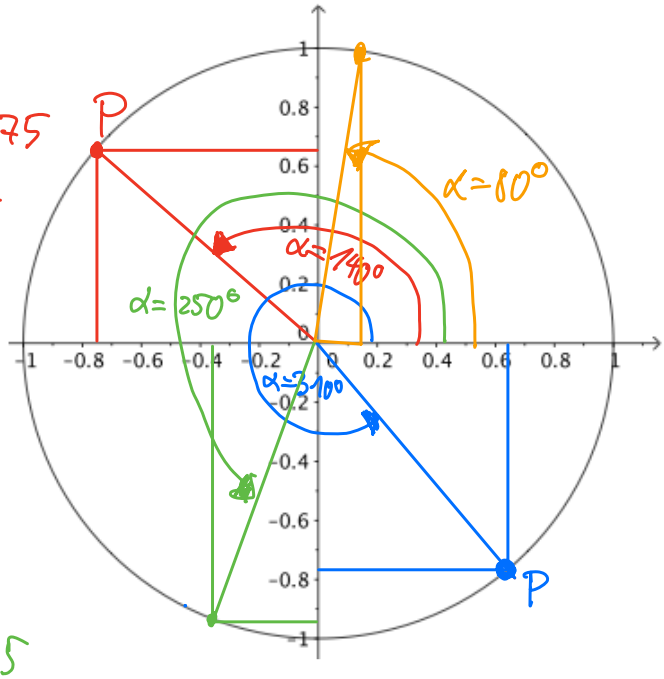
Dies nutzt man zur Verallgemeinerung für beliebige Winkel $\alpha \in \mathbb{R}$:

$x_P = \cos(\alpha), \quad y_P = \sin(\alpha)$

Ü7.4

a) Bild $\Rightarrow$ $x_P = \cos(140) \approx -0{,}75$

$y_P = \sin(140) \approx 0.65$



b) $x_P = \cos(310) \approx 0.64$

$y_P = \sin(310) \approx -0.76$

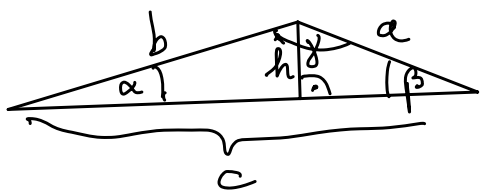
c) $x_P = \cos(250) \approx -0.35$

$y_P = \sin(250) \approx -0.95$

d) $x_P = \cos(80) \approx 0.15$

$y_P = \sin(80) \approx 0.98$

Sinussatz und Cosinussatz im allgemeinen Dreieck:



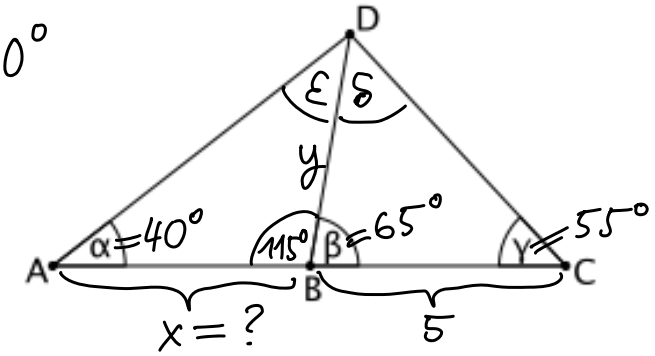
$$\sin(\alpha) = \frac{h}{b}, \quad \sin(\beta) = \frac{h}{a}$$

$$\Rightarrow \quad b \cdot \sin(\alpha) = a \cdot \sin(\beta)$$

Sinussatz: $$\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$$

(Ist α oder $\beta > 90°$, benötigt es ein paar Zusatzüberlegungen in der Herleitung.)

Ü 7.5 $\delta = 180 - 65 - 55 = 60^\circ$



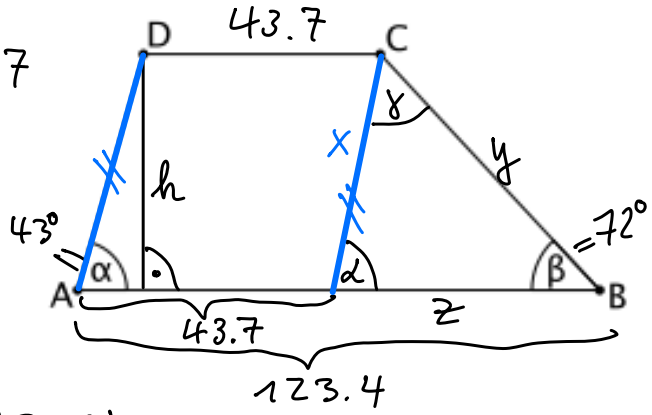
$$\frac{\sin(55)}{y} = \frac{\sin(60)}{5}$$

$$\Rightarrow y = 4.73$$

$$\varepsilon = 180 - 40 - 115 = 25^\circ$$

$$\frac{\sin(\varepsilon)}{x} = \frac{\sin(40)}{y} \Rightarrow x = 3.11$$

Ü 7.6 $z = 123.4 - 43.7 = 79.7$



$$\gamma = 180 - \alpha - \beta = 65^\circ$$

$$\frac{\sin(\alpha)}{y} = \frac{\sin(\gamma)}{z} \Rightarrow y = 59.97$$

$$\frac{\sin(\beta)}{x} = \frac{\sin(\gamma)}{z} \Rightarrow x = 83.64$$

$$\Rightarrow \text{Umfang} = 123.4 + 43.7 + y + x = 310.71$$

$$\sin(\alpha) = \frac{h}{x} \Rightarrow h = 57.03$$

$$\Rightarrow \text{Fläche} = 43.7 \cdot h + \frac{1}{2} z h = 4764.86$$